



COS 730 – Assignment 1

Advanced Behavioural Modelling and Responsibility-Driven Design

Due date: 19th March 2026 @ 11:00 am

Background

Software systems often fail not because functionality is missing, but because interactions, responsibilities, and system behaviour are poorly designed. In this assignment, you will analyse a system scenario, model object interactions, apply responsibility assignment principles, and optimise your modelling approach to improve design clarity and efficiency.

You are required to evaluate multiple use cases, construct behavioural models, apply responsibility assignment patterns, and justify how your modelling approach improves development efficiency and maintainability.

Instructions

You are required to complete the following tasks and submit a PDF report. Include screenshots of all modelling with explanations and justifications. Please make use of diagrams.net or modelling tools, using paint or hand drawn modelling will attract 0 marks.

Scenario

A university is developing an **Intelligent Research Collaboration Platform** that assists researchers in managing projects, datasets, publications, and collaborative activities.

The system includes features such as:

- Research project management
- Dataset submission and validation
- Automated reviewer assignment
- Research collaboration tracking
- Research output evaluation

You are provided with **five core** use cases below.

Use Cases

- **Use Case 1: Create Research Project**

A researcher creates a new research project and defines collaborators, objectives, and milestones.

System requirements include validating project information, assigning project roles, recording project metadata, notifying collaborators.

- **Use Case 2: Submit Dataset**

A researcher uploads a dataset associated with a project.

System requirements include validate dataset format, scan for integrity issues, store dataset metadata, link dataset to a research project.

- **Use Case 3: Automated Reviewer Assignment**

When a research output is submitted, the system automatically assigns reviewers.

System requirements include identify suitable reviewers, avoid conflicts of interest, balance reviewer workload, record assignment decisions.

- **Use Case 4: Collaboration Monitoring**

The platform monitors collaboration activity within projects.

System requirements include track interactions between researchers, detect inactive collaborations, generate collaboration insights.

- **Use Case 5: Research Output Evaluation**

A submitted research output is evaluated by reviewers and scored.

System requirements include collect review scores, aggregate evaluation results, identify consensus or disagreement, produce evaluation reports.

Tasks

For any 3 of the use cases above complete the tasks below:

Task 1 – Interaction Modelling [30 marks]

1. Develop a **Sequence Diagram** showing object interactions.
2. Identify:
 - Participating objects
 - Message flows
 - Control responsibilities
3. Highlight decision points and system responses.
You must also:
 - Identify modelling redundancies across use cases.
 - Propose a **refined interaction modelling approach** that reduces duplication.

Task 2 – Responsibility Assignment Analysis [15 marks]

Using responsibility assignment principles (e.g., GRASP), you must:

1. Assign responsibilities to system components.
2. Justify your design choices.

3. Evaluate:

- Coupling
- Cohesion
- Scalability of responsibilities

You must demonstrate how your design **improves system modularity**.

Task 3 – Modelling System Behaviour [15 marks]

Identify which of the use cases represent **specialised system types**, such as:

- Event-driven systems
- Rule-based decision systems
- Data-processing or transformational systems

Then:

1. Explain the modelling challenges involved.
2. Propose improved modelling strategies.
3. Adapt your models accordingly.

Task 4 – Design Optimisation [40 marks]

Analyse how modelling can be **optimised across the system**.

Discuss:

- Model reuse strategies
- Shared interaction patterns
- Generalised controllers or services
- Reduction of modelling complexity

You must explain:

- How improved modelling reduces development effort
- How it supports maintainability
- How it improves system evolution

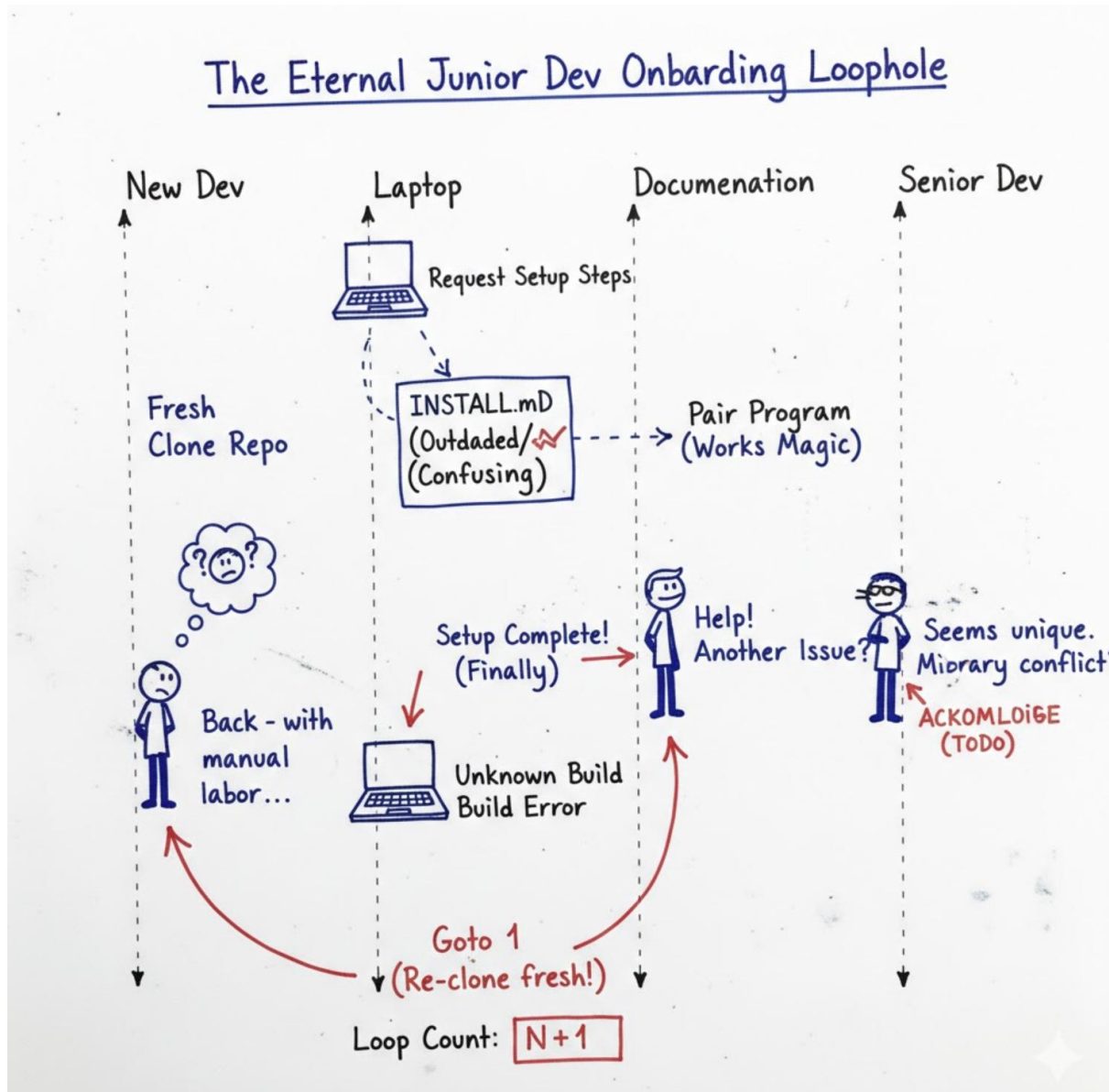
Support your analysis with revised diagrams or design improvements.

Submission

Submit your report as a PDF file ClickUP before the deadline.

Please upload well in advance, as there will be

ABSOLUTELY NO LATE SUBMISSIONS!!!



P.S. This is AI generated, so at least you are safe doing sequence diagrams in this assignment 😊

TOTAL: 100